1 Introduction

GEB 2508 AI & Society

Content

- Artificial Intelligence
  - Machine Learning: Mechanism & Limitations
  - Development & Applications
- Science and Technology Studies (STS)
  - Science & Technology: Intellectual History
  - Social and Political Implications
- Course Overview

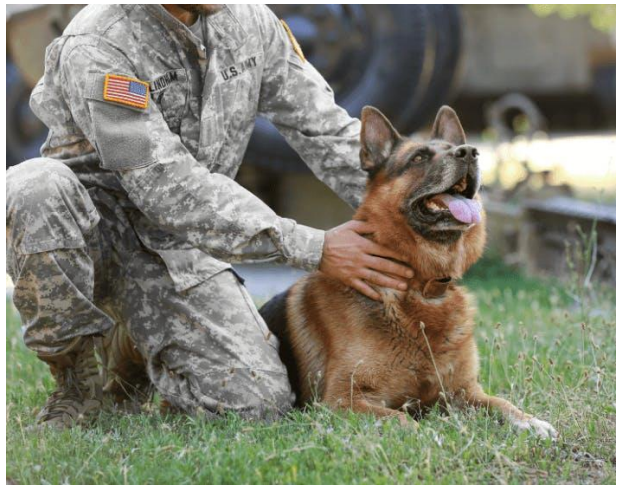
What is Learning?

How do humans learn?

- People learn by experience: watching, practicing, making mistakes, and improving.

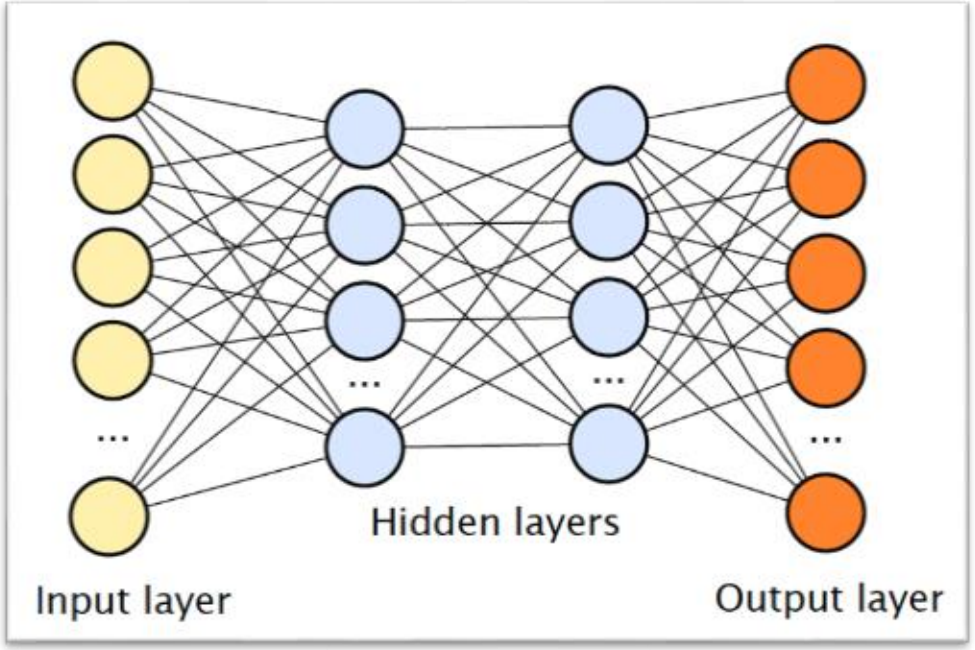
How do animals learn?

- Animals learn through trial and error and **rewards**.



What is Machine Learning?

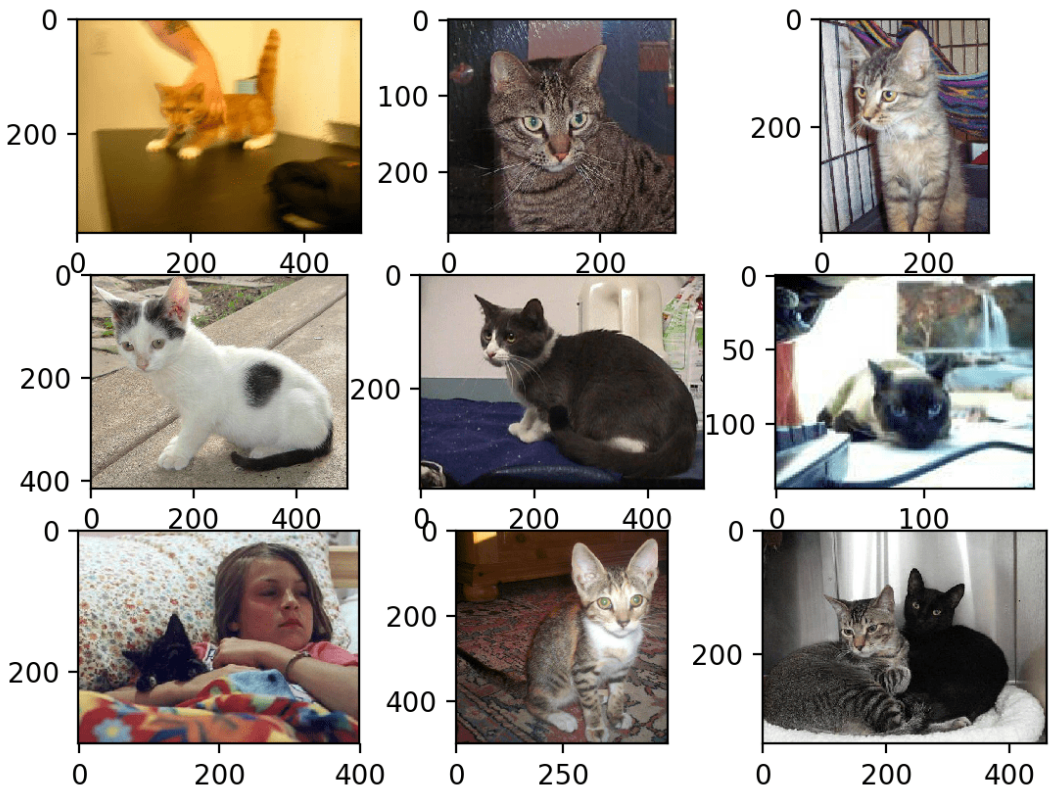
- A way of teaching machines to find patterns in data
- Neural networks: multiple layers of artificial “neurons” that process information hierarchically
  - Simplified model of the human brain
  - Interconnected nodes (neurons)
  - Signals flow from inputs → hidden layers → outputs



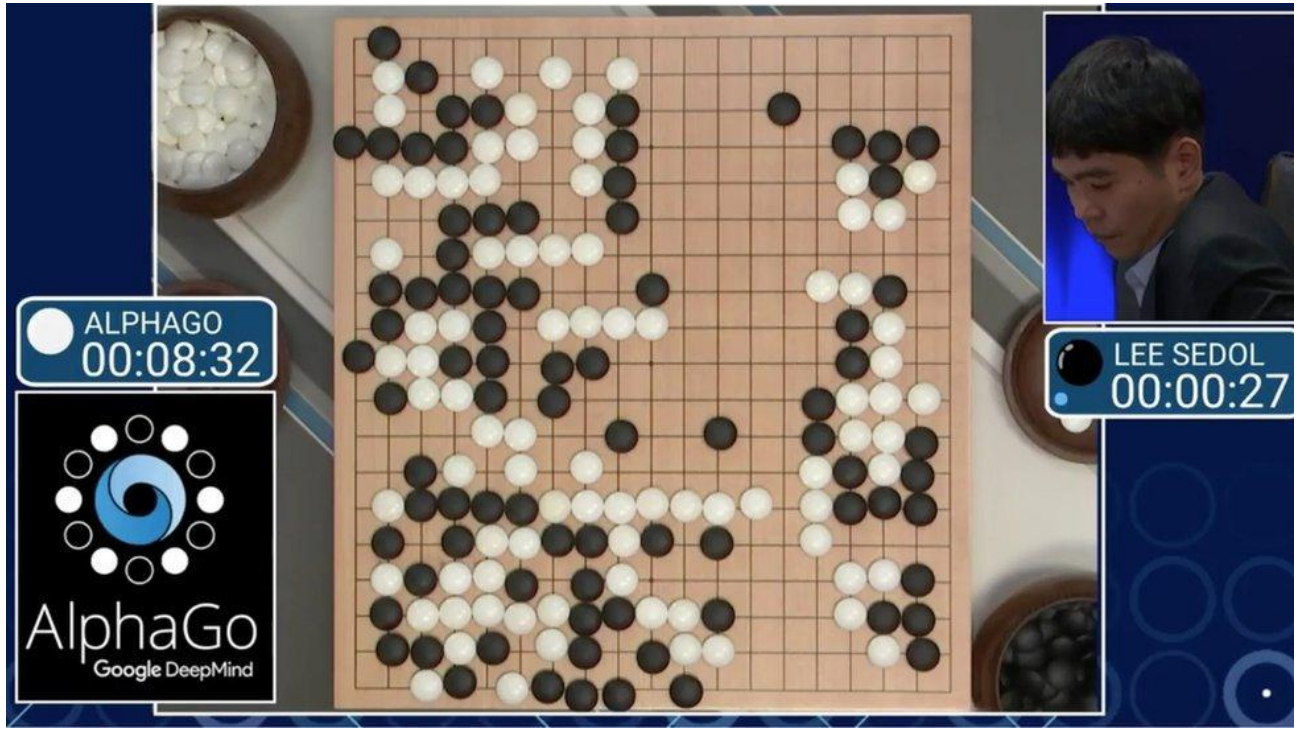
Neural networks in deep learning

How Does AI Learn?

- Traditional programming: **explicit** instructions and rules written by humans
  - E.g., “cats have pointy ears, four legs, a tail…”
- Deep learning: rules discovered **automatically** from data.
  - Learns from examples, instead of following hand-written rules

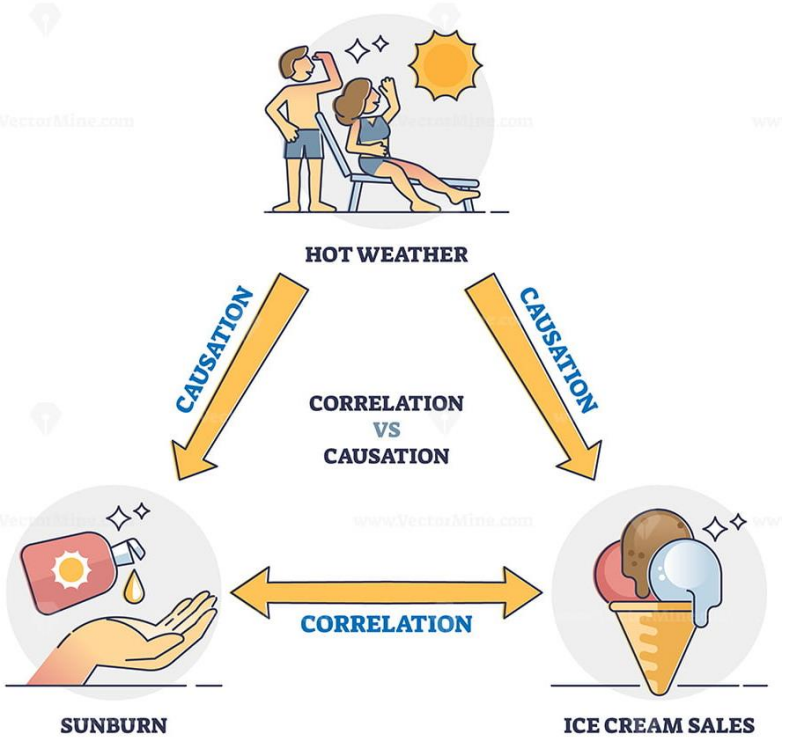


Examples from Games



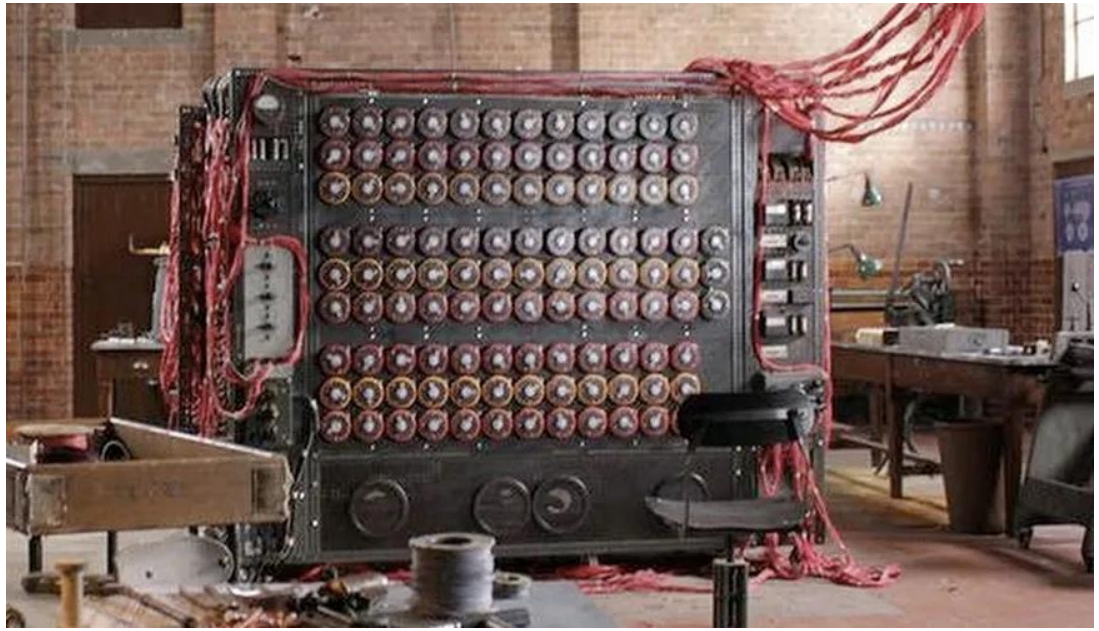
Limits of Machine Learning

- Data hungry: requires huge labeled datasets
- Weak generalization: good on training data but struggles with new situations
- No causal understanding: detects **correlations** but doesn't know **causes**
- Poor abstract reasoning: limited in logic and commonsense knowledge
- Opaque: works like a black box, hard to justify decisions



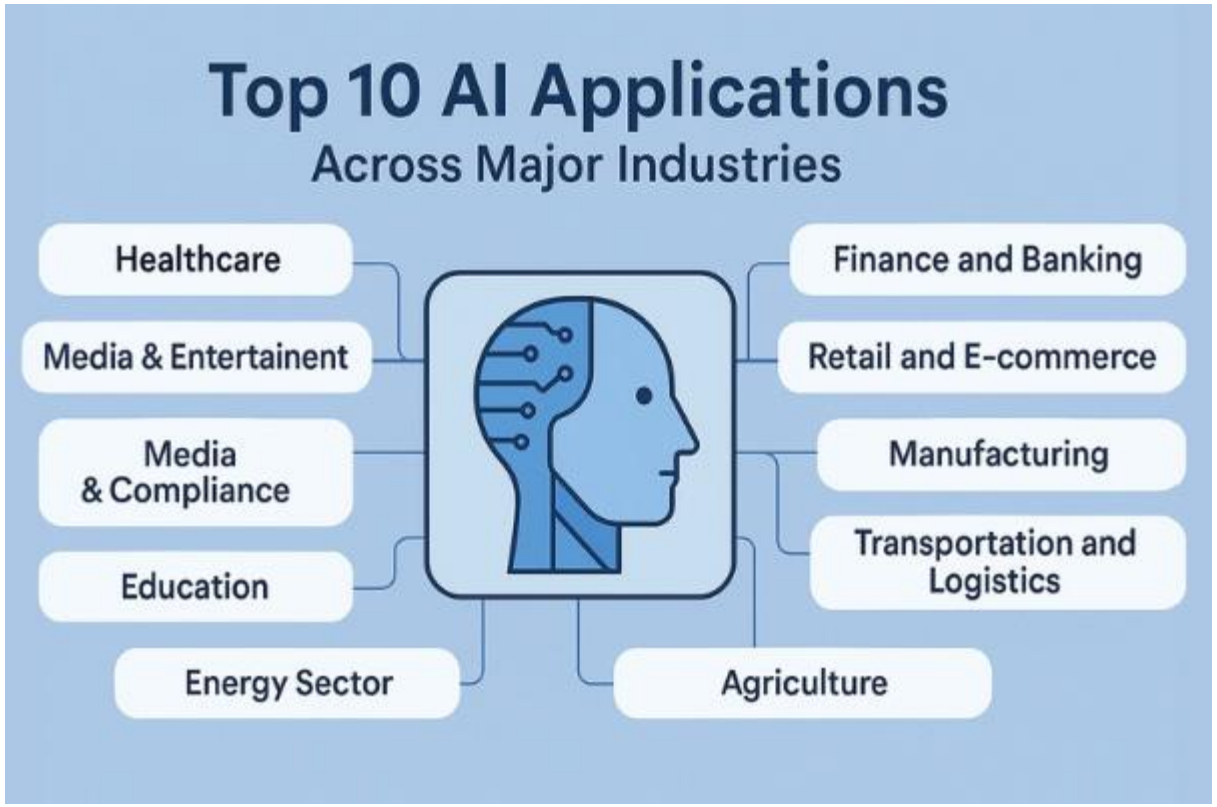
Historical Development

- Long intellectual history of the idea of a Turing machine
- Why now? Three enabling factors:
  - **Big data** (millions of photos, recordings, text)
  - **Powerful hardware** (GPUs)
  - **Better algorithms** (e.g., backpropagation refinements)



Practical Applications

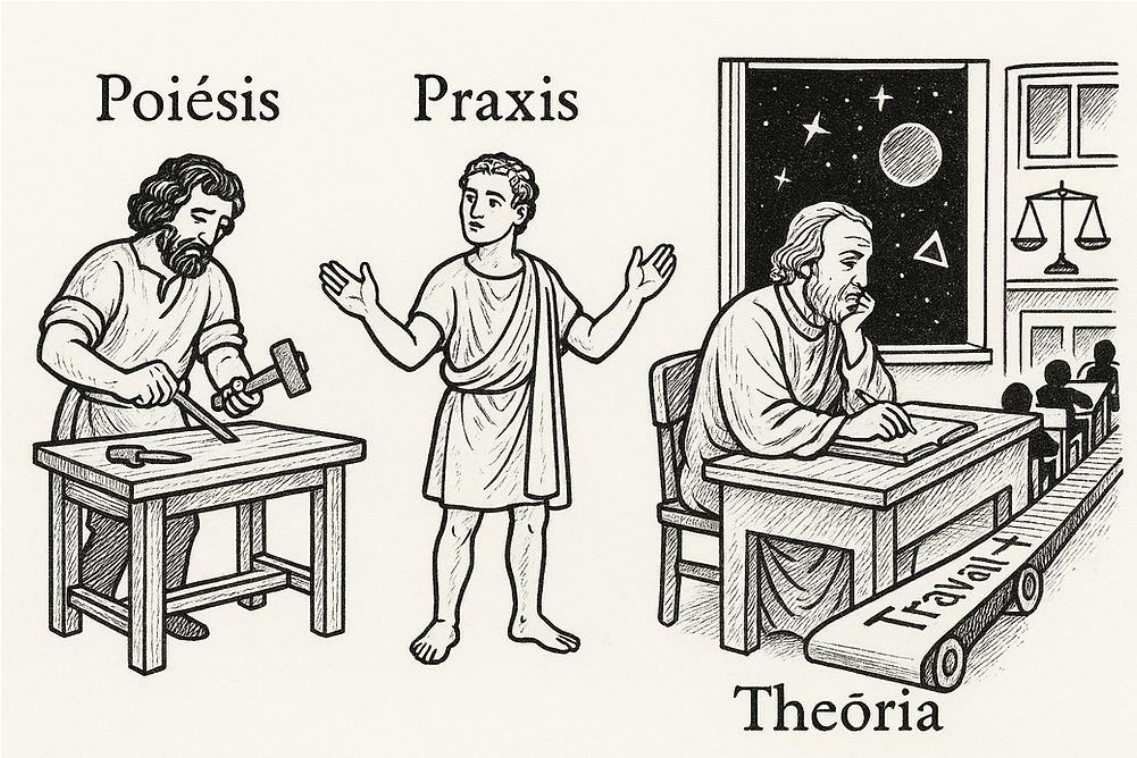
- Everyday Life
  - Voice assistants
  - Image recognition
  - Language translation
  - Text generation
  - Self-driving cars
- Health & Medicine
- Science
- Arts & Recreation



Three Types of Knowledge

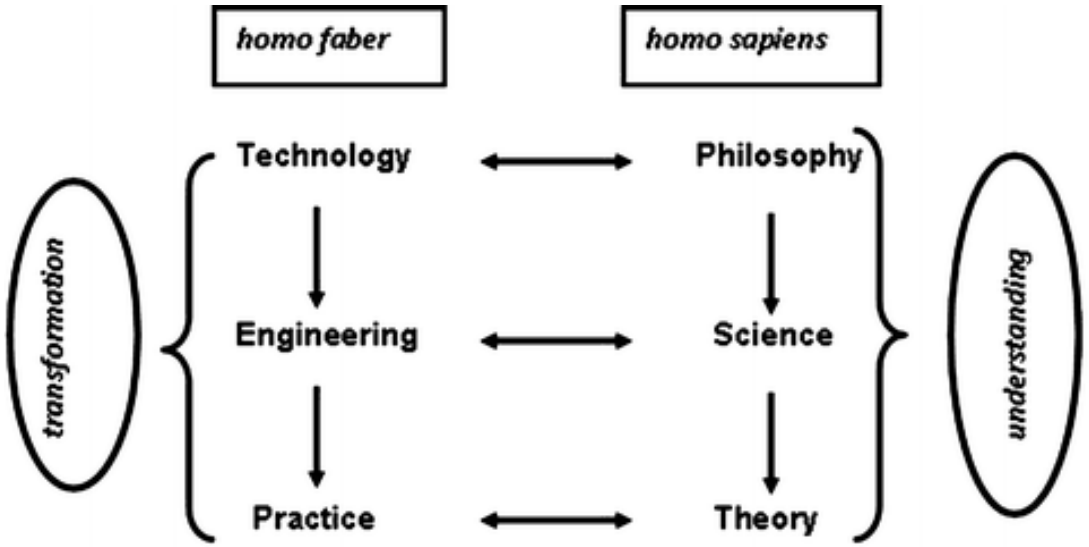
- Distinguished by Aristotle in *Nicomachean Ethics*

1. Theoretical Knowledge (Epistēmē - Theōria)
2. Productive Knowledge (Technē - Praxis)
3. Practical Wisdom (Phronēsis - Poiesis)



Homo Sapiens vs. Homo Faber

- Homo sapiens (the knowing human) aligns with *theōria* and *sophia* (theoretical wisdom)
- Homo faber (the making human) aligns with *poiesis* and *technē* (productive skill)
- Homo sapiens emphasizes knowing **why**, while homo faber emphasizes knowing **how**.

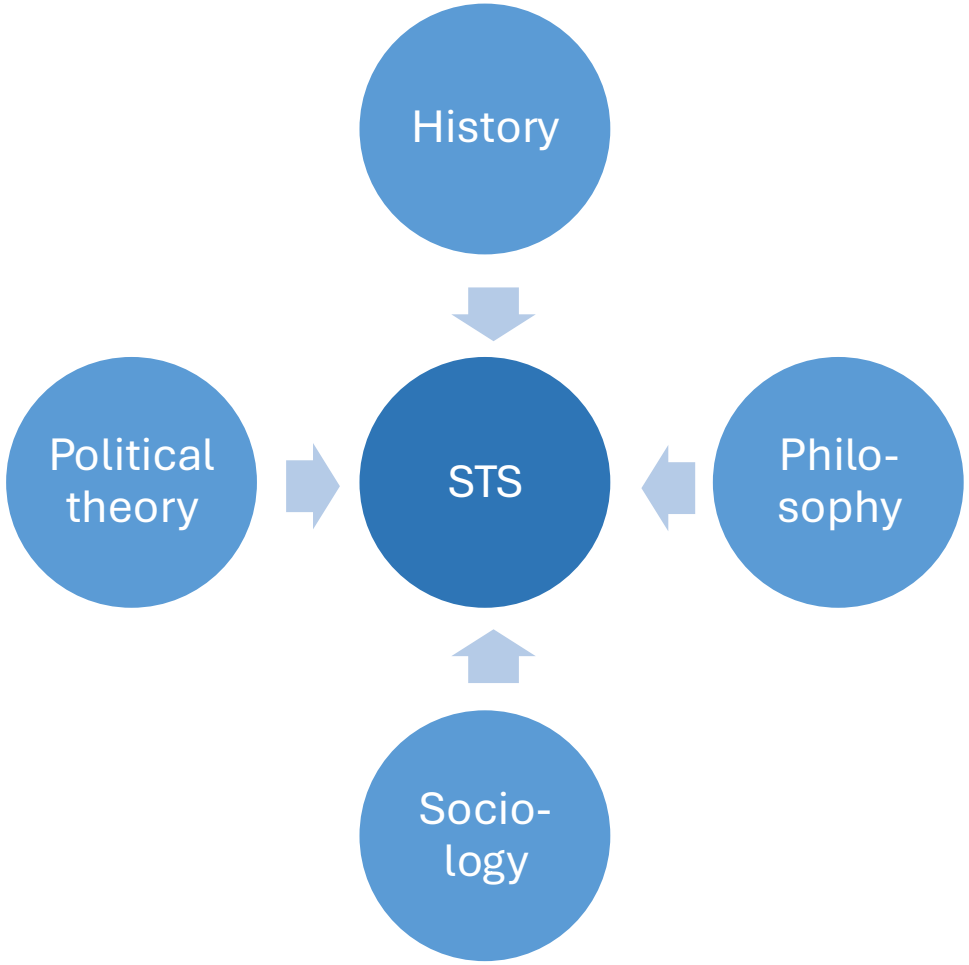


Science and Technology Studies (STS)

- The field of STS emerged in the mid-20th century.
  - Increasing impact of science and technology in modern society
  - Social movements criticize the role of science and technology
- Science and technology are deeply embedded in socio-political contexts.
- Social implications of science and technology: informed decision-making

Interdisciplinary Field

- History: science & technology as **products** of specific times
- Philosophy: explores how **values** enter "neutral" inquiry; raises ethical questions
- Sociology: treats science & technology as **collective** social practices
- Political theory: highlights **power**, governance, democracy



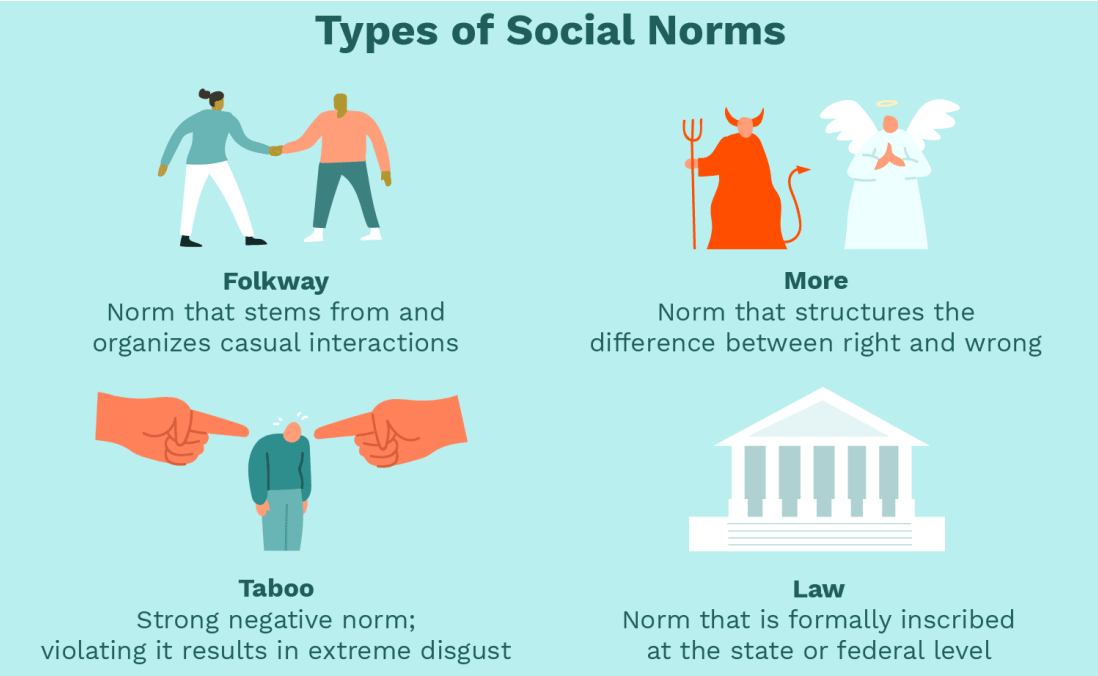
STS Programs and Institutions

- Program on Science, Technology and Society, Harvard University
- Program in Science, Technology, and Society, MIT
- Center for Science, Technology, Medicine and Society (CSTMS), UC Berkeley
- Department of Science & Technology Studies, Cornell University
- Program in Science, Technology, and Society, Stanford University

- 清华大学 科学技术与社会研究所
- 中国科学院 自然科学史研究所 科技与社会研究中心
- 中国社会科学院 科学技术和社会研究中心
- 中国科学技术大学 科技哲学系 科学、技术与社会的关系
- 南方科技大学 社会科学高等研究院 科技与社会研究
- STS minor program at CUHK(SZ)

What Makes Something Social

- Society is the result of **people** and **institutions** interacting with one another. Society in turn shapes the people and institutions that form it.
- The effects of society operate through the vague mechanism of social **norms**. They are not rational - or rather, their rationality is not universal.



S & T Are Social

- In the simplest sense, S&T are the products of people.
- Social context shapes scientific and technological development.
- Scientific knowledge and technological artifacts are **socially constructed**.
- S&T have profound social impacts.

What Makes AI Social

- Collective production: organizational decisions and negotiations
- Social norms and professional practices: standards of "good" AI **vary** across contexts
- Historical and cultural context: training data **reflect** existing social patterns
- Use and interpretation: users adapt, misuse, and reinterpret AI

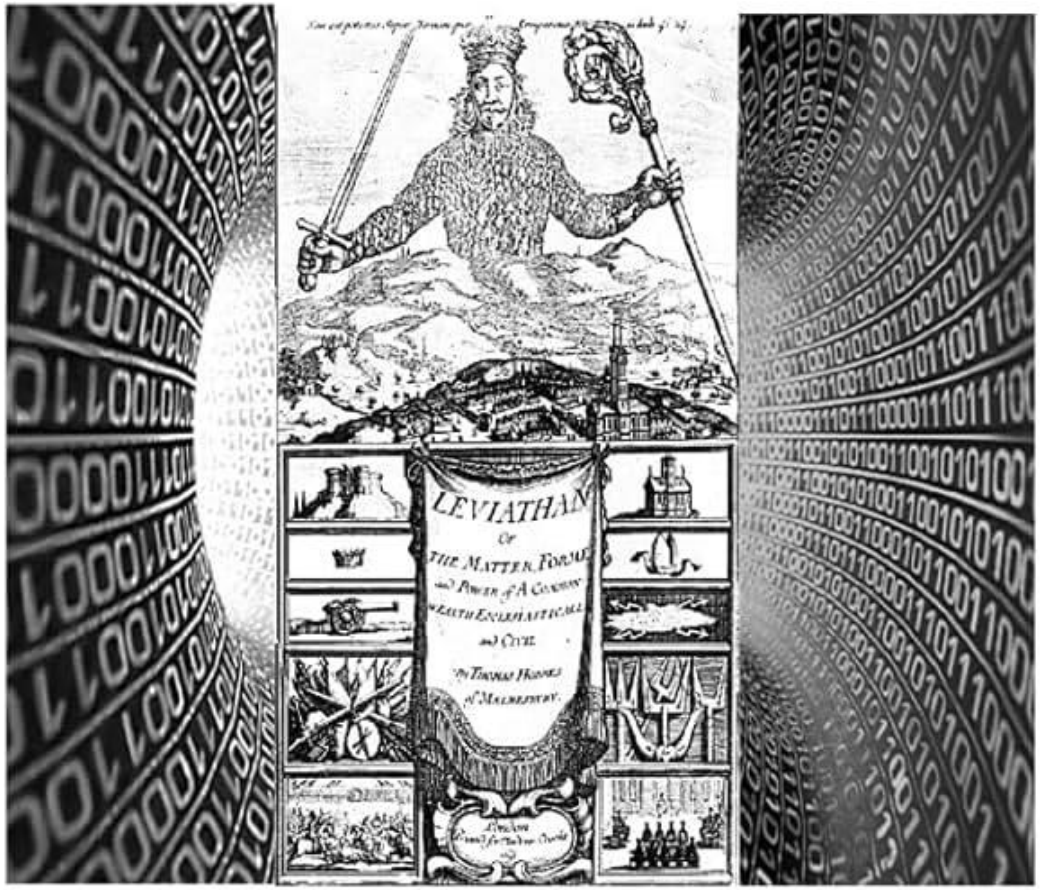
A contestant was asked in the 2024 Miss Korea pageant: "If the deepfake version of myself is more attractive, how should I bridge the gap between the deepfake version and the real me?"

What Makes Something Political

- Politics is about **power**: how it is distributed, exercised, and contested within societies.
- Politics operates through formal mechanisms as well as informal, invisible mechanisms, such as **discourses** (what can be said and what cannot be said).
- Political activity functions to wield and **consolidate** power.

S & T Are Political

- The development and use of technologies are regulated by governments, with decisions often shaped by political ideologies and agendas.
- Scientific communities seek to advance their political goals through S&T.
- S&T as **tools** of power and control: redistribution, inclusion and exclusion



What Makes AI Political

- Funding and priority setting
- **Concentration** of resources and power
- Authority and expertise
  - Reinforcing existing inequalities while **appearing objective**
- Classifying and ranking people

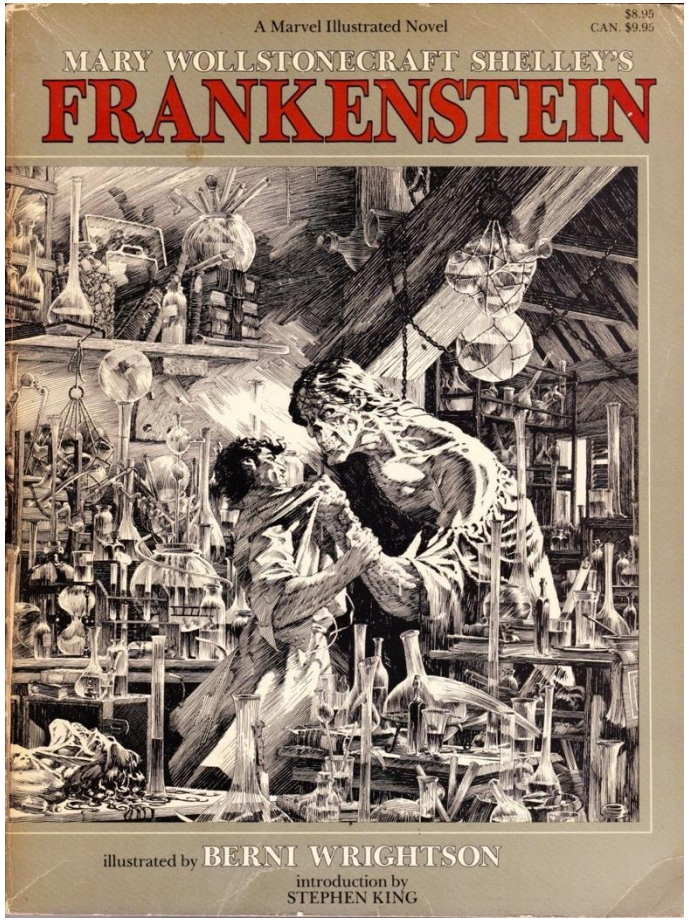
Murdered Insurance CEO Had Deployed an AI to Automatically Deny Benefits for Sick People

Just over a year before United Healthcare CEO Brian Thompson was murdered this week in Midtown Manhattan, a lawsuit filed against the insurance giant he helmed revealed just how draconian its claims-denying process had become.

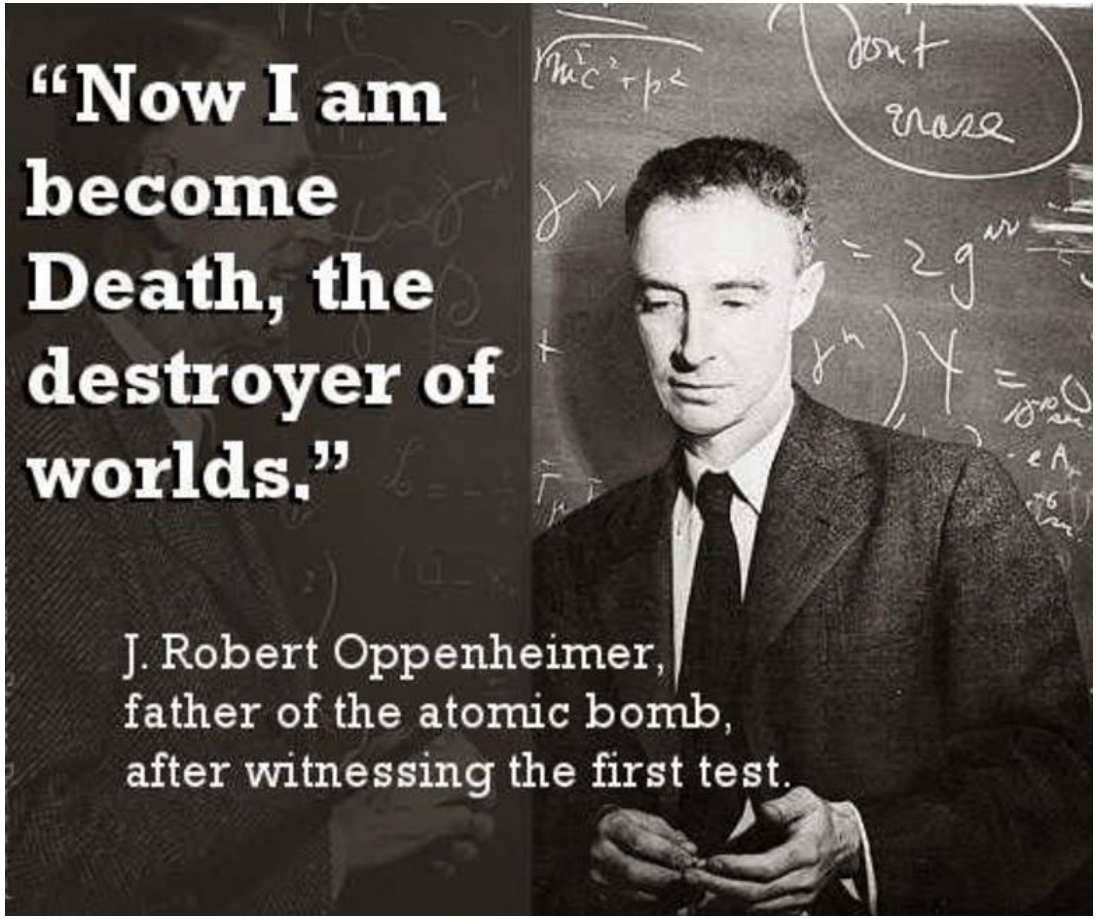
Last November, the estates of two former UHC patients filed suit in Minnesota alleging that the insurer used an AI algorithm to deny and override claims to elderly patients that had been approved by their doctors.

The algorithm in question, known as nH Predict, allegedly had a 90 percent error rate — and according to the families of the two deceased men who filed the suit, UHC knew it.

Pandora’s Box & Frankenstein



The Case of Oppenheimer



Course Schedule

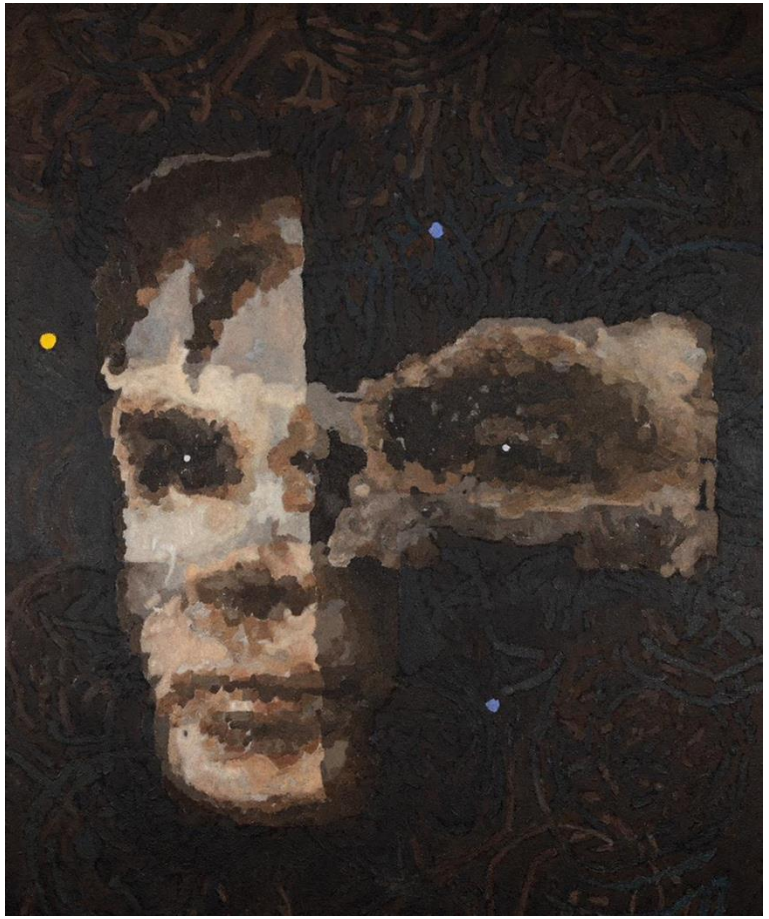
Section	Topic
/	Intro
I. Technology and Society: Theoretical Perspectives	Social Constructivism: the Making of Technology
	Marxian Perspectives: Labor, Capital, and Automation
	Foucauldian Perspectives: Power, Governmentality, and the Panopticon
	Feminist Perspectives: Rights, Representation, and Situated Knowledge
	Postcolonial Perspectives: Empire, Racialization, and Global Infrastructure
II. AI and Society: Contemporary Issues	Human–Machine Relations: Structure-Agency, Cyborgs, and Socio-Technical Systems
	AI and Politics: Warfare, Geopolitics, and Industrial Policy
	AI and Intellectual Property: Knowledge, Commons, and Privatization
	AI and Environment: Malthusian Limits, Cornucopianism, and Sustainability
	AI and Ethics: Algorithmic Bias, Fairness, and Accountability
/	Final Exam
	Paper Consultation

Class Policy

- Late Work Policy
  - Documentation is necessary
- Attendance
  - Request for leave must be made **before** class with supporting documentation
  - Three unexcused absences will lead to academic failure
- Academic Integrity
  - **Zero** t**olerance** on academic dishonesty

Assessment

- Participation and Engagement: 20%
  - In-class participation and online participation
- Presentation: 20%
  - Explore creative applications of AI
- Term Paper: 30%
  - Analyze power and knowledge production in AI systems
- Final Exam: 30%
  - The exam will be open-book but electronic devices are prohibited; ensure that all required readings are completed before the exam

Experiential Activity: AI News of the Year

- Choose one AI-related news from the past year that you deem particularly impactful, inspiring, controversial, or consequential.
- Share it in our class group and explain why you think it is significant within the context of AI-society interactions.